Q1) Three slot machines with the following observed rewards.

Machine	observed Rewards
A	3, 2, 4
B	6, 7
C	5, 4, 6

formula:

$$Q(a) = \frac{\text{sum of rewards}}{N(a)}$$

Therefore;

Machine A = $Q(A) = \frac{3+2+4}{3}$

$Q(A) = \frac{9}{3}$

$\boxed{Q(A) = 3}$

Machine B = $Q(B) = \frac{6+7}{2}$

$= \frac{13}{2}$

$\boxed{Q(B) = 6.5}$

Machine C = $Q(C) = \frac{5+4+6}{3}$

$= \frac{15}{3}$

$\boxed{Q(C) = 5}$

Greedy selection:-

with highest $Q(a)$

$Q(A) = 3$

$Q(B) = 6.5$

$Q(C) = 5$

Therefore,

$\max(3, 6.5, 5) = 6.5$

Hence,

Greedy selected Machine = B

Q2) The Incremental Update formula:-

$$Q_{new}(a) = Q_{old}(a) + \frac{R - Q_{old}(a)}{N(a)}$$

Initial values:

$Q(A) = Q(B) = Q(C) = 0$

Machine A

observed reward $R = 3$

Increment count $N(A) = 1$

$$Q(A) = 0 + \frac{3-0}{1}$$

Current Values:-

$Q(A) = 3, \quad Q(B) = 0, \quad Q(C) = 0$

$R = 2$

$N(A) = 2$

$Q_{old}(A) = 3$

$$Q(A) = 3 + \frac{2-3}{2}$$

$$= 3 - \frac{1}{2}$$

$$\boxed{Q(A) = 2.5}$$

$Q(A) = 2.5 \quad Q(B) = 0 \quad Q(C) = 0$

$R = 4$

$N(A) = 3$

$$Q(A) = 2.5 + \frac{4-2.5}{3}$$

$$= 2.5 + \frac{1.5}{3}$$

$$= 2.5 + 0.5$$

$$Q(A) = 3$$

$Q(A) = 3, \quad Q(B) = 0, \quad Q(C) = 0$

Machine B:

$R = 6$

$N_{old}(B) = 0$

$N(B) = 1$

$$Q(B) = 0 + \frac{6-0}{1}$$

$$Q(B) = 6$$

$$\boxed{Q(B) = 6}$$

$R = 7$

$N(B) = 2$

$$Q(B) = 6 + \frac{7-6}{2}$$

$$= 6 + \frac{1}{2}$$

$$= 6 + 0.5$$

$$\boxed{Q(B) = 6.5}$$

Machine C

$R = 5$

$N(C) = 1$

$$Q(C) = 0 + \frac{5-0}{1}$$

$$\boxed{Q(C) = 5}$$

$R = 4$

$N(C) = 2$

$$Q(C) = 5 + \frac{4-5}{2}$$

$$= 5 + \frac{-1}{2}$$

$$= 5 - 0.5$$

$$\boxed{Q(C) = 4.5}$$

$R = 6$

$N(C) = 3$

$$Q(C) = 4.5 + \frac{6-4.5}{3}$$

$$= 4.5 + 0.5$$

$$\boxed{Q(C) = 5}$$

Final Increment Values

Machine	Reward Sequence	Increment values	Final Q(a)
A	3, 2, 4	$3 \to 2.5 \to 3$	3.0
B	6, 7	$6 \to 6.5$	6.5
C	5, 4, 6	$5 \to 4.5 \to 5$	5.0

Highest value is $\boxed{Q(B) = 6.5}$,